

Reducción de dimensión: PC - Proyecto modulo 6

Eumir Rendon, Raúl Llamosas, etc

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```
# library(learnr)
library(tidyverse)

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.3      v tibble     3.3.1
## v lubridate  1.9.5      v tidyr      1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

library(tidymodels)

## -- Attaching packages ----- tidymodels 1.5.0 --
## v broom       1.0.13      v rsample     1.3.2
## v dials       1.4.3      v tailor      0.1.0
## v infer       1.1.0      v tune        2.1.0
## v modeldata   1.5.1      v workflows   1.3.0
## v parsnip     1.6.0      v workflowsets 1.1.1
## v recipes     1.3.3      v yardstick   1.4.0
## -- Conflicts ----- tidymodels_conflicts() --
## x scales::discard() masks purrr::discard()
## x dplyr::filter()   masks stats::filter()
## x recipes::fixed()  masks stringr::fixed()
## x dplyr::lag()      masks stats::lag()
## x yardstick::spec() masks readr::spec()
## x recipes::step()   masks stats::step()

library(embed)
library(corr)
library(tidytext)
library(gradethis)
library(sortable)

## To use sortable with shiny modules, run `sortable::enable_modules()` to opt into the new standard. S

library(learntidymodels)
library(rstatix)

##
## Adjuntando el paquete: 'rstatix'
##
```

```

## The following objects are masked from 'package:infer':
##
##   chisq_test, prop_test, t_test
##
## The following object is masked from 'package:dials':
##
##   get_n
##
## The following object is masked from 'package:stats':
##
##   filter
library(broom)
library(rgl)
library(plotly)

##
## Adjuntando el paquete: 'plotly'
##
## The following object is masked from 'package:ggplot2':
##
##   last_plot
##
## The following object is masked from 'package:stats':
##
##   filter
##
## The following object is masked from 'package:graphics':
##
##   layout
library(GGally)
library(tidyr)
library(FactoMineR)
library(factoextra)

## Welcome to factoextra!
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practice
library(ggord)
library(modeldata)
library(tsne)
library(uwot)

## Cargando paquete requerido: Matrix
##
## Adjuntando el paquete: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##   expand, pack, unpack
library(scattermore)
library(readr)
library(Rtsne)
library(gganimate)
library(data.table)

```

```
##
## Adjuntando el paquete: 'data.table'
##
## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
##     week, yday, year
##
## The following objects are masked from 'package:dplyr':
##
##     between, first, last
##
## The following object is masked from 'package:purrr':
##
##     transpose
library(baguette)
library(discrim)

##
## Adjuntando el paquete: 'discrim'
##
## The following object is masked from 'package:dials':
##
##     smoothness
theme_set(theme_bw(16))

tidymodels_prefer()

df<-read.csv("/home/yamorza/Documentos/espacio_trabajo_intro_analisis_dat2/proyecto_modulo_6/src/habits
```

Comenzaremos con nuestra reduccion PCA

Vamos a estandarizar nuestras variables y lidiar con las variables categóricas mencionadas en el paso anterior por medio de dummies.

```
df_pca <- df %>%
  select(-exam_score)

receta_pca <- recipe(~ ., data = df_pca) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_predictors())

datos_preparados <- prep(receta_pca) %>%
  bake(new_data = NULL)

datos_finales <- datos_preparados %>%
  bind_cols(exam_score = df$exam_score)

print("Datos preparados para PCA:")

## [1] "Datos preparados para PCA:"
```

```
print(head(datos_finales))
```

```
## # A tibble: 6 x 20
##   age study_hours_per_day social_media_hours netflix_hours
##   <dbl>         <dbl>         <dbl>         <dbl>
## 1  1.08         -2.42         -1.11         -0.669
## 2 -0.216         2.28          0.251          0.447
## 3  0.217        -1.46          0.507         -0.483
## 4  1.08        -1.74          1.19         -0.762
## 5 -0.649         0.987         1.62         -1.23
## 6  1.52          2.48         -1.03         -1.69
## # i 16 more variables: attendance_percentage <dbl>, sleep_hours <dbl>,
## #   exercise_frequency <dbl>, mental_health_rating <dbl>, gender_Male <dbl>,
## #   gender_Other <dbl>, part_time_job_Yes <dbl>, diet_quality_Good <dbl>,
## #   diet_quality_Poor <dbl>, parental_education_level_High.School <dbl>,
## #   parental_education_level_Master <dbl>, parental_education_level_None <dbl>,
## #   internet_quality_Good <dbl>, internet_quality_Poor <dbl>,
## #   extracurricular_participation_Yes <dbl>, exam_score <dbl>
```

```
print("\nEstructura de los datos:")
```

```
## [1] "\nEstructura de los datos:"
```

```
str(datos_finales)
```

```
## tibble [1,000 x 20] (S3: tbl_df/tbl/data.frame)
##  $ age                        : num [1:1000] 1.084 -0.216 0.217 1.084 -0.649 ...
##  $ study_hours_per_day        : num [1:1000] -2.417 2.281 -1.464 -1.736 0.987 ...
##  $ social_media_hours         : num [1:1000] -1.114 0.251 0.507 1.189 1.616 ...
##  $ netflix_hours              : num [1:1000] -0.669 0.447 -0.483 -0.762 -1.227 ...
##  $ attendance_percentage      : num [1:1000] 0.0924 1.401 1.135 -1.3971 0.7201 ...
##  $ sleep_hours                : num [1:1000] 1.25 -1.52 1.25 2.23 -1.28 ...
##  $ exercise_frequency         : num [1:1000] 1.4604 1.4604 -1.0082 0.473 -0.0207 ...
##  $ mental_health_rating       : num [1:1000] 0.9 0.9 -1.56 -1.56 -1.56 ...
##  $ gender_Male                : num [1:1000] -0.955 -0.955 1.047 -0.955 -0.955 ...
##  $ gender_Other               : num [1:1000] -0.209 -0.209 -0.209 -0.209 -0.209 ...
##  $ part_time_job_Yes          : num [1:1000] -0.523 -0.523 -0.523 -0.523 -0.523 ...
##  $ diet_quality_Good          : num [1:1000] -0.779 1.282 -0.779 -0.779 -0.779 ...
##  $ diet_quality_Poor          : num [1:1000] -0.476 -0.476 2.098 2.098 -0.476 ...
##  $ parental_education_level_High.School: num [1:1000] -0.803 1.245 1.245 -0.803 -0.803 ...
##  $ parental_education_level_Master    : num [1:1000] 2.232 -0.448 -0.448 2.232 2.232 ...
##  $ parental_education_level_None      : num [1:1000] -0.316 -0.316 -0.316 -0.316 -0.316 ...
##  $ internet_quality_Good              : num [1:1000] -0.899 -0.899 -0.899 1.112 1.112 ...
##  $ internet_quality_Poor              : num [1:1000] -0.439 -0.439 2.273 -0.439 -0.439 ...
##  $ extracurricular_participation_Yes  : num [1:1000] 1.464 -0.683 -0.683 1.464 -0.683 ...
##  $ exam_score                        : num [1:1000] 56.2 100 34.3 26.8 66.4 100 89.8 72.6 78.9 100
```

```
print("\nResumen estadístico:")
```

```
## [1] "\nResumen estadístico:"
```

```
summary(datos_finales)
```

```
##      age      study_hours_per_day social_media_hours netflix_hours
##  Min.   :-1.5155   Min.   :-2.41686   Min.   :-2.137028   Min.   :-1.69256
##  1st Qu.: -0.7573   1st Qu.: -0.64682   1st Qu.: -0.687039   1st Qu.: -0.76243
##  Median :-0.2158   Median :-0.03411   Median :-0.004691   Median :-0.01832
```

```

## Mean : 0.0000 Mean : 0.00000 Mean : 0.000000 Mean : 0.00000
## 3rd Qu.: 1.0840 3rd Qu.: 0.64668 3rd Qu.: 0.677657 3rd Qu.: 0.65602
## Max. : 1.5173 Max. : 3.23367 Max. : 4.004103 Max. : 3.33015
## attendance_percentage sleep_hours exercise_frequency
## Min. : -2.99297 Min. : -2.66647 Min. : -1.50191
## 1st Qu.: -0.65236 1st Qu.: -0.70949 1st Qu.: -1.00818
## Median : 0.02854 Median : 0.02438 Median : -0.02074
## Mean : 0.00000 Mean : 0.00000 Mean : 0.00000
## 3rd Qu.: 0.73339 3rd Qu.: 0.67671 3rd Qu.: 0.96671
## Max. : 1.68825 Max. : 2.87832 Max. : 1.46044
## mental_health_rating gender_Male gender_Other part_time_job_Yes
## Min. : -1.5586 Min. : -0.9545 Min. : -0.2093 Min. : -0.5231
## 1st Qu.: -0.8562 1st Qu.: -0.9545 1st Qu.: -0.2093 1st Qu.: -0.5231
## Median : -0.1538 Median : -0.9545 Median : -0.2093 Median : -0.5231
## Mean : 0.0000 Mean : 0.0000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 0.8997 3rd Qu.: 1.0466 3rd Qu.: -0.2093 3rd Qu.: -0.5231
## Max. : 1.6021 Max. : 1.0466 Max. : 4.7735 Max. : 1.9098
## diet_quality_Good diet_quality_Poor parental_education_level_High.School
## Min. : -0.7792 Min. : -0.4762 Min. : -0.8026
## 1st Qu.: -0.7792 1st Qu.: -0.4762 1st Qu.: -0.8026
## Median : -0.7792 Median : -0.4762 Median : -0.8026
## Mean : 0.0000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 1.2821 3rd Qu.: -0.4762 3rd Qu.: 1.2448
## Max. : 1.2821 Max. : 2.0979 Max. : 1.2448
## parental_education_level_Master parental_education_level_None
## Min. : -0.4475 Min. : -0.3162
## 1st Qu.: -0.4475 1st Qu.: -0.3162
## Median : -0.4475 Median : -0.3162
## Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: -0.4475 3rd Qu.: -0.3162
## Max. : 2.2323 Max. : 3.1590
## internet_quality_Good internet_quality_Poor extracurricular_participation_Yes
## Min. : -0.8986 Min. : -0.4395 Min. : -0.6825
## 1st Qu.: -0.8986 1st Qu.: -0.4395 1st Qu.: -0.6825
## Median : -0.8986 Median : -0.4395 Median : -0.6825
## Mean : 0.0000 Mean : 0.0000 Mean : 0.0000
## 3rd Qu.: 1.1117 3rd Qu.: -0.4395 3rd Qu.: 1.4637
## Max. : 1.1117 Max. : 2.2733 Max. : 1.4637
## exam_score
## Min. : 18.40
## 1st Qu.: 58.48
## Median : 70.50
## Mean : 69.60
## 3rd Qu.: 81.33
## Max. : 100.00

```

```
print("\nNombres de las columnas:")
```

```
## [1] "\nNombres de las columnas:"
```

```
names(datos_finales)
```

```
## [1] "age"
```

```
## [2] "study_hours_per_day"
```

```
## [3] "social_media_hours"
```

```
## [4] "netflix_hours"
## [5] "attendance_percentage"
## [6] "sleep_hours"
## [7] "exercise_frequency"
## [8] "mental_health_rating"
## [9] "gender_Male"
## [10] "gender_Other"
## [11] "part_time_job_Yes"
## [12] "diet_quality_Good"
## [13] "diet_quality_Poor"
## [14] "parental_education_level_High.School"
## [15] "parental_education_level_Master"
## [16] "parental_education_level_None"
## [17] "internet_quality_Good"
## [18] "internet_quality_Poor"
## [19] "extracurricular_participation_Yes"
## [20] "exam_score"

pca_modelo <- prcomp(datos_preparados, center = FALSE, scale. = FALSE)

varianza_explicada <- summary(pca_modelo)$importance

print("Varianza explicada por cada componente:")

## [1] "Varianza explicada por cada componente:"

print(varianza_explicada)

##           PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation  1.22255 1.214206 1.159799 1.112669 1.066908 1.039401
## Proportion of Variance 0.07866 0.077590 0.070800 0.065160 0.059910 0.056860
## Cumulative Proportion 0.07866 0.156260 0.227060 0.292220 0.352130 0.408990
##           PC7      PC8      PC9      PC10     PC11     PC12
## Standard deviation  1.025275 1.021238 1.004379 1.001977 0.9942167 0.9839586
## Proportion of Variance 0.055330 0.054890 0.053090 0.052840 0.0520200 0.0509600
## Cumulative Proportion 0.464310 0.519200 0.572300 0.625140 0.6771600 0.7281200
##           PC13     PC14     PC15     PC16     PC17
## Standard deviation  0.9708561 0.9627759 0.929615 0.8835386 0.7810519
## Proportion of Variance 0.0496100 0.0487900 0.045480 0.0410900 0.0321100
## Cumulative Proportion 0.7777300 0.8265100 0.872000 0.9130800 0.9451900
##           PC18     PC19
## Standard deviation  0.758339 0.68288
## Proportion of Variance 0.030270 0.02454
## Cumulative Proportion 0.975460 1.00000

print("\nProporción de varianza acumulada:")

## [1] "\nProporción de varianza acumulada:"

print(varianza_explicada[3, 1:10])

##      PC1      PC2      PC3      PC4      PC5      PC6      PC7      PC8      PC9      PC10
## 0.07866 0.15626 0.22706 0.29222 0.35213 0.40899 0.46431 0.51920 0.57230 0.62514

num_componentes <- length(pca_modelo$sdev)

scree_data <- data.frame(
```

```

Componente = 1:num_componentes,
Varianza = pca_modelo$sdev^2 / sum(pca_modelo$sdev^2) * 100,
Varianza_Acumulada = cumsum(pca_modelo$sdev^2 / sum(pca_modelo$sdev^2)) * 100
)

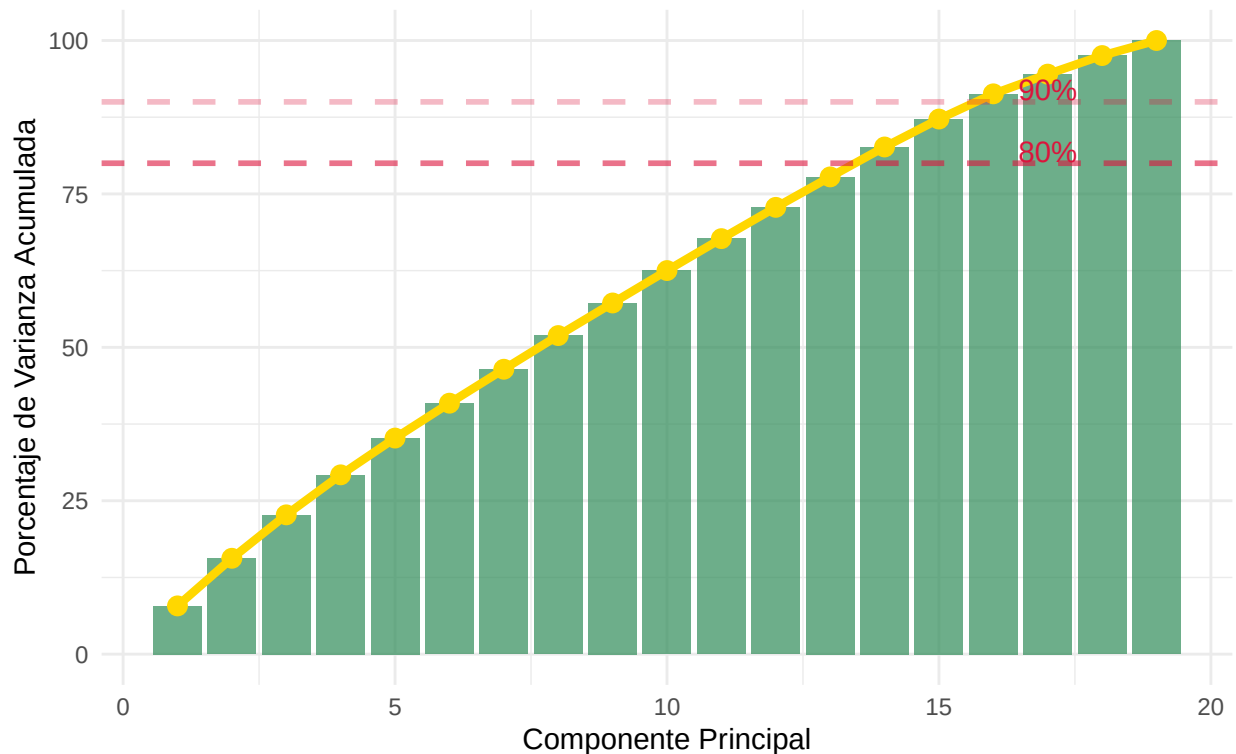
ggplot(scrree_data, aes(x = Componente)) +
  geom_col(aes(y = Varianza_Acumulada), fill = "#2E8B57", alpha = 0.7) +
  geom_line(aes(y = Varianza_Acumulada), color = "#FFD700", size = 1.5) +
  geom_point(aes(y = Varianza_Acumulada), color = "#FFD700", size = 3) +
  geom_hline(yintercept = 80, linetype = "dashed", color = "#DC143C", alpha = 0.6, size = 1) +
  geom_hline(yintercept = 90, linetype = "dashed", color = "#DC143C", alpha = 0.3, size = 1) +
  annotate("text", x = num_componentes - 2, y = 82, label = "80%", color = "#DC143C", size = 4) +
  annotate("text", x = num_componentes - 2, y = 92, label = "90%", color = "#DC143C", size = 4) +
  labs(x = "Componente Principal",
       y = "Porcentaje de Varianza Acumulada",
       title = "Scree Plot - Varianza Acumulada",
       subtitle = "Barras y línea muestran la varianza acumulada") +
  theme_minimal() +
  theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14),
        plot.subtitle = element_text(color = "#4682B4", size = 10))

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

Scree Plot – Varianza Acumulada

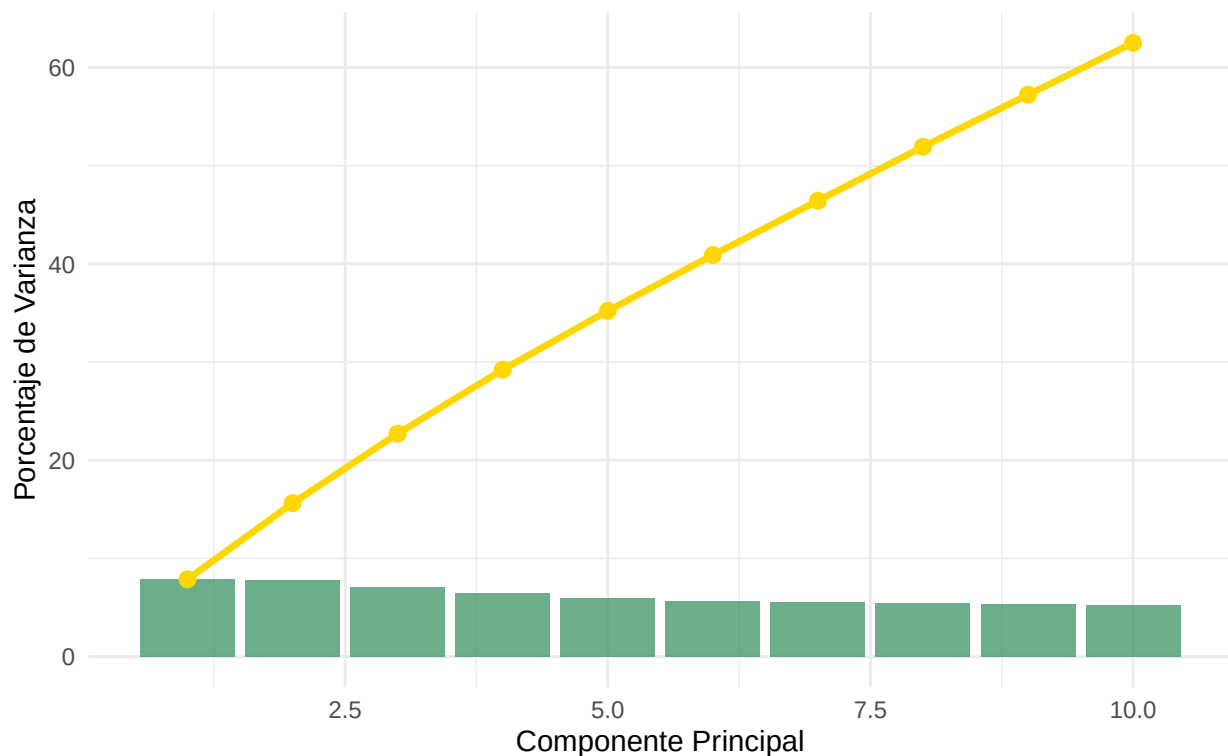
Barras y línea muestran la varianza acumulada



```
ggplot(scree_data[1:10, ], aes(x = Componente)) +
  geom_col(aes(y = Varianza), fill = "#2E8B57", alpha = 0.7) +
  geom_line(aes(y = Varianza_Acumulada), color = "#FFD700", size = 1.2) +
  geom_point(aes(y = Varianza_Acumulada), color = "#FFD700", size = 2.5) +
  labs(x = "Componente Principal",
       y = "Porcentaje de Varianza",
       title = "Scree Plot - Varianza Individual y Acumulada (primeros 10 componentes)",
       subtitle = "Barras: varianza individual | Línea: varianza acumulada") +
  theme_minimal() +
  theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14),
        plot.subtitle = element_text(color = "#4682B4", size = 10))
```

Scree Plot – Varianza Individual y Acumulada (primeros 10 compo

Barras: varianza individual | Línea: varianza acumulada



Notamos que necesitamos muchas componentes para explicar varianza. ¿Qué pasa si eliminamos las variables categóricas? Como vimos en el análisis exploratorio de datos, no tienen mucha correlación con exam score. Reducir el número de columnas definitivamente nos ayudaría en simplificar el modelo.

```
df_sin_categoricas <- df %>%
  select(study_hours_per_day, mental_health_rating, exercise_frequency,
         sleep_hours, attendance_percentage, netflix_hours, social_media_hours, exam_score)
```

```
df_pca <- df_sin_categoricas %>%
  select(-exam_score)
```

```
receta_pca <- recipe(~ ., data = df_pca) %>%
  step_dummy(all_nominal_predictors()) %>%
  step_normalize(all_predictors())
```



```
datos_preparados <- prep(receta_pca) %>%
  bake(new_data = NULL)
```

```
datos_finales <- datos_preparados %>%
  bind_cols(exam_score = df$exam_score)
```

```
print("Datos preparados para PCA:")
```

```
## [1] "Datos preparados para PCA:"
```

```
print(head(datos_finales))
```

```
## # A tibble: 6 x 8
##   study_hours_per_day mental_health_rating exercise_frequency sleep_hours
##           <dbl>           <dbl>           <dbl>           <dbl>
## 1          -2.42             0.900             1.46             1.25
## 2           2.28             0.900             1.46            -1.52
## 3          -1.46            -1.56            -1.01             1.25
## 4          -1.74            -1.56             0.473            2.23
## 5           0.987            -1.56            -0.0207           -1.28
## 6           2.48            -0.505            -1.01             0.758
## # i 4 more variables: attendance_percentage <dbl>, netflix_hours <dbl>,
## #   social_media_hours <dbl>, exam_score <dbl>
```

```
print("\nEstructura de los datos:")
```

```
## [1] "\nEstructura de los datos:"
```

```
str(datos_finales)
```

```
## tibble [1,000 x 8] (S3: tbl_df/tbl/data.frame)
## $ study_hours_per_day : num [1:1000] -2.417 2.281 -1.464 -1.736 0.987 ...
## $ mental_health_rating : num [1:1000] 0.9 0.9 -1.56 -1.56 -1.56 ...
## $ exercise_frequency : num [1:1000] 1.4604 1.4604 -1.0082 0.473 -0.0207 ...
## $ sleep_hours : num [1:1000] 1.25 -1.52 1.25 2.23 -1.28 ...
## $ attendance_percentage: num [1:1000] 0.0924 1.401 1.135 -1.3971 0.7201 ...
## $ netflix_hours : num [1:1000] -0.669 0.447 -0.483 -0.762 -1.227 ...
## $ social_media_hours : num [1:1000] -1.114 0.251 0.507 1.189 1.616 ...
## $ exam_score : num [1:1000] 56.2 100 34.3 26.8 66.4 100 89.8 72.6 78.9 100 ...
```

```
print("\nResumen estadístico:")
```

```
## [1] "\nResumen estadístico:"
```

```
summary(datos_finales)
```

```
##   study_hours_per_day mental_health_rating exercise_frequency sleep_hours
##   Min.   : -2.41686   Min.   : -1.5586   Min.   : -1.50191   Min.   : -2.66647
##   1st Qu.: -0.64682   1st Qu.: -0.8562   1st Qu.: -1.00818   1st Qu.: -0.70949
##   Median : -0.03411   Median : -0.1538   Median : -0.02074   Median :  0.02438
##   Mean   :  0.00000   Mean   :  0.0000   Mean   :  0.00000   Mean   :  0.00000
##   3rd Qu.:  0.64668   3rd Qu.:  0.8997   3rd Qu.:  0.96671   3rd Qu.:  0.67671
##   Max.   :  3.23367   Max.   :  1.6021   Max.   :  1.46044   Max.   :  2.87832
##   attendance_percentage netflix_hours social_media_hours exam_score
##   Min.   : -2.99297   Min.   : -1.69256   Min.   : -2.137028   Min.   : 18.40
##   1st Qu.: -0.65236   1st Qu.: -0.76243   1st Qu.: -0.687039   1st Qu.: 58.48
##   Median :  0.02854   Median : -0.01832   Median : -0.004691   Median : 70.50
```

```

## Mean      : 0.00000      Mean      : 0.00000      Mean      : 0.000000      Mean      : 69.60
## 3rd Qu.: 0.73339      3rd Qu.: 0.65602      3rd Qu.: 0.677657      3rd Qu.: 81.33
## Max.      : 1.68825      Max.      : 3.33015      Max.      : 4.004103      Max.      :100.00

print("\nNombres de las columnas:")

## [1] "\nNombres de las columnas:"

names(datos_finales)

## [1] "study_hours_per_day"      "mental_health_rating"      "exercise_frequency"
## [4] "sleep_hours"              "attendance_percentage"      "netflix_hours"
## [7] "social_media_hours"      "exam_score"

pca_modelo <- prcomp(datos_preparados, center = FALSE, scale. = FALSE)

varianza_explicada <- summary(pca_modelo)$importance

print("Varianza explicada por cada componente:")

## [1] "Varianza explicada por cada componente:"

print(varianza_explicada)

##
##          PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation  1.040938 1.020002 1.012393 0.9965108 0.986644 0.9714435
## Proportion of Variance 0.154790 0.148630 0.146420 0.1418600 0.139070 0.1348100
## Cumulative Proportion 0.154790 0.303420 0.449840 0.5917000 0.730770 0.8655900
##
##          PC7
## Standard deviation  0.9700004
## Proportion of Variance 0.1344100
## Cumulative Proportion 1.0000000

print("\nProporción de varianza acumulada:")

## [1] "\nProporción de varianza acumulada:"

print(varianza_explicada)

##
##          PC1      PC2      PC3      PC4      PC5      PC6
## Standard deviation  1.040938 1.020002 1.012393 0.9965108 0.986644 0.9714435
## Proportion of Variance 0.154790 0.148630 0.146420 0.1418600 0.139070 0.1348100
## Cumulative Proportion 0.154790 0.303420 0.449840 0.5917000 0.730770 0.8655900
##
##          PC7
## Standard deviation  0.9700004
## Proportion of Variance 0.1344100
## Cumulative Proportion 1.0000000

num_componentes <- length(pca_modelo$sdev)

scree_data <- data.frame(
  Componente = 1:num_componentes,
  Varianza = pca_modelo$sdev^2 / sum(pca_modelo$sdev^2) * 100,
  Varianza_Acumulada = cumsum(pca_modelo$sdev^2 / sum(pca_modelo$sdev^2)) * 100
)

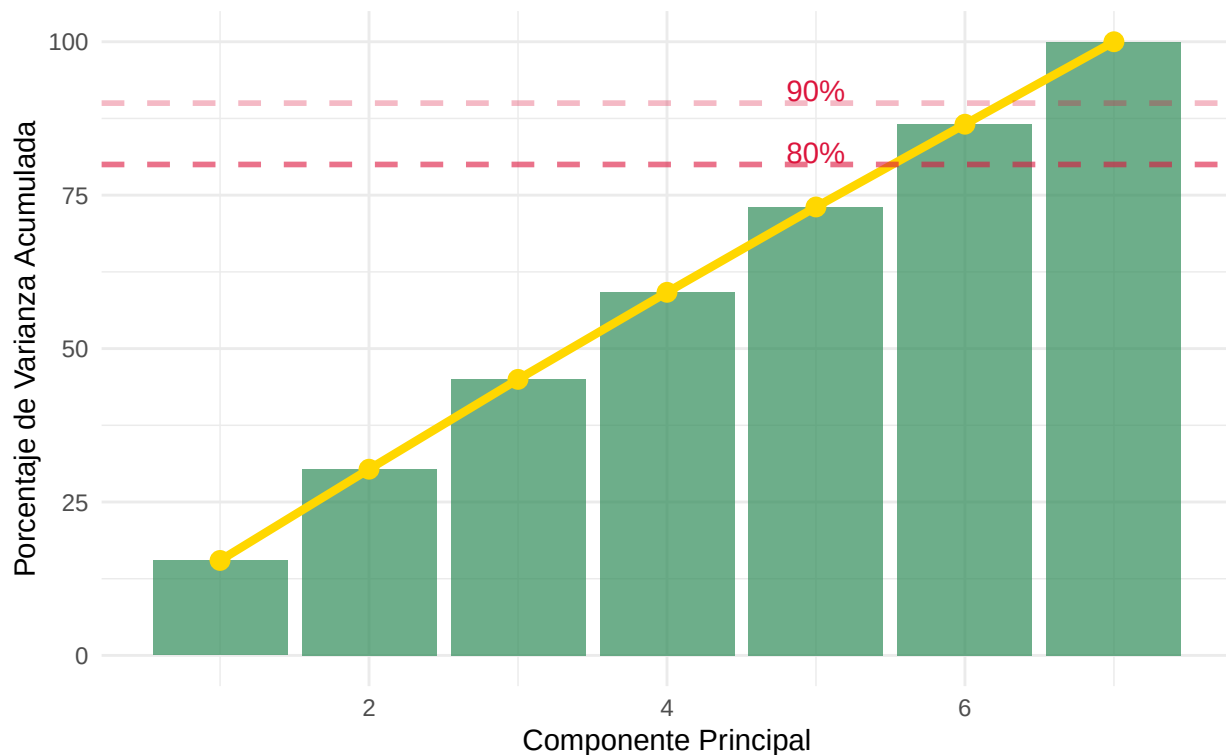
ggplot(scree_data, aes(x = Componente)) +
  geom_col(aes(y = Varianza_Acumulada), fill = "#2E8B57", alpha = 0.7) +
  geom_line(aes(y = Varianza_Acumulada), color = "#FFD700", size = 1.5) +

```

```
geom_point(aes(y = Varianza_Acumulada), color = "#FFD700", size = 3) +
geom_hline(yintercept = 80, linetype = "dashed", color = "#DC143C", alpha = 0.6, size = 1) +
geom_hline(yintercept = 90, linetype = "dashed", color = "#DC143C", alpha = 0.3, size = 1) +
annotate("text", x = num_componentes - 2, y = 82, label = "80%", color = "#DC143C", size = 4) +
annotate("text", x = num_componentes - 2, y = 92, label = "90%", color = "#DC143C", size = 4) +
labs(x = "Componente Principal",
     y = "Porcentaje de Varianza Acumulada",
     title = "Scree Plot - Varianza Acumulada",
     subtitle = "Barras y línea muestran la varianza acumulada") +
theme_minimal() +
theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14),
      plot.subtitle = element_text(color = "#4682B4", size = 10))
```

Scree Plot – Varianza Acumulada

Barras y línea muestran la varianza acumulada



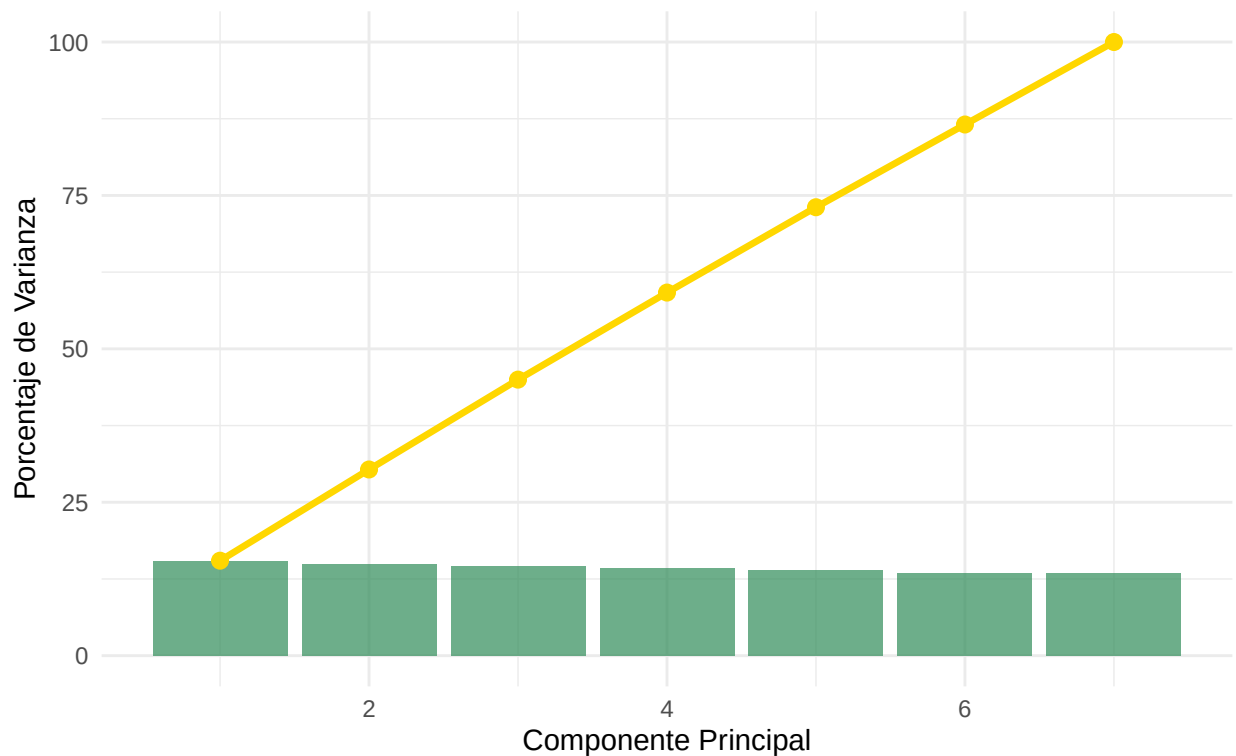
```
ggplot(scrree_data[1:10, ], aes(x = Componente)) +
geom_col(aes(y = Varianza), fill = "#2E8B57", alpha = 0.7) +
geom_line(aes(y = Varianza_Acumulada), color = "#FFD700", size = 1.2) +
geom_point(aes(y = Varianza_Acumulada), color = "#FFD700", size = 2.5) +
labs(x = "Componente Principal",
     y = "Porcentaje de Varianza",
     title = "Scree Plot - Varianza Individual y Acumulada (primeros 10 componentes)",
     subtitle = "Barras: varianza individual | Línea: varianza acumulada") +
theme_minimal() +
theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14),
      plot.subtitle = element_text(color = "#4682B4", size = 10))
```

Warning: Removed 3 rows containing missing values or values outside the scale range

```
## (`geom_col()`).
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_line()`).
## Warning: Removed 3 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Scree Plot – Varianza Individual y Acumulada (primeros 10 comp

Barras: varianza individual | Línea: varianza acumulada



Como podemos ver, solo necesitamos 5 componentes para explicar una varianza cercana a 80%. Esto es una clara mejoría con respecto de mantener todas las columnas.

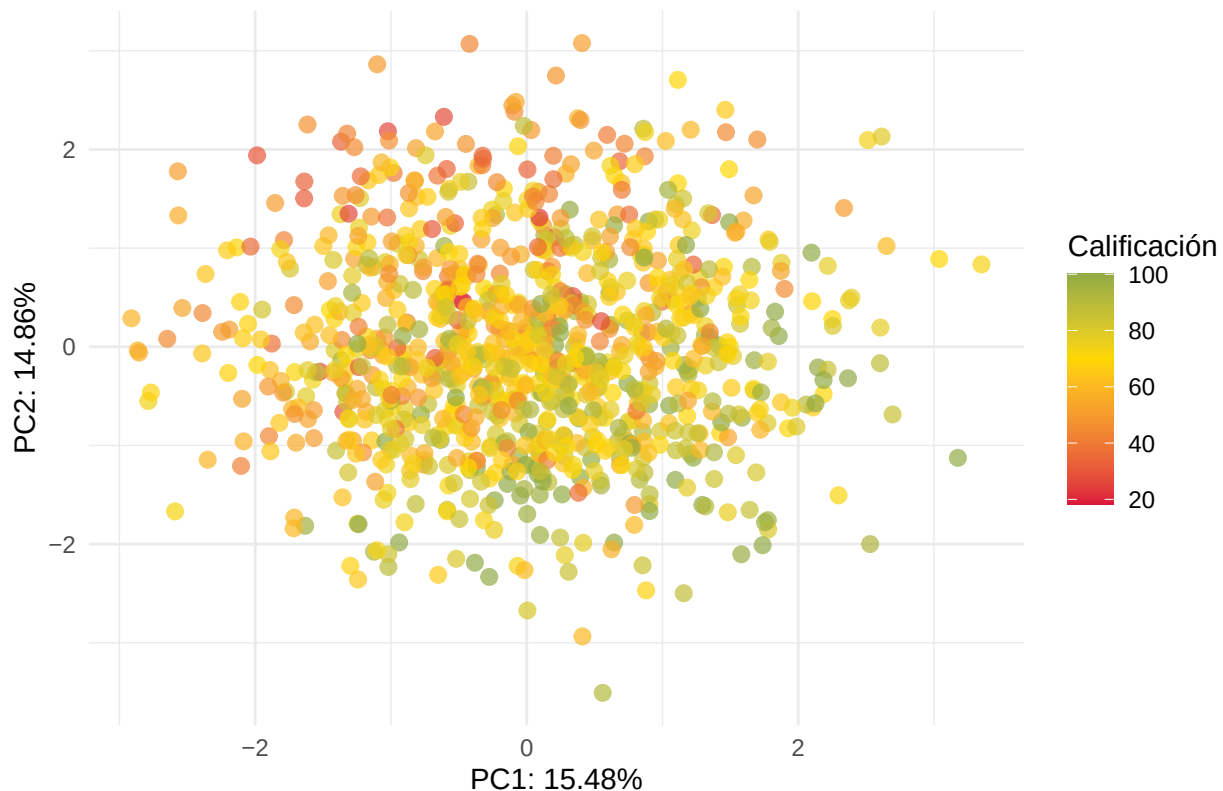
```
scores_pca <- as.data.frame(pca_modelo$x)

scores_pca <- scores_pca %>%
  bind_cols(exam_score = df$exam_score)

ggplot(scores_pca, aes(x = PC1, y = PC2, color = exam_score)) +
  geom_point(size = 2.5, alpha = 0.7) +
  scale_color_gradient2(
    low = "#DC143C",
    mid = "#FFD700",
    high = "#2E8B57",
    midpoint = 70,
    name = "Calificación"
  ) +
  labs(x = paste0("PC1: ", round(varianza_explicada[2, 1] * 100, 2), "%"),
       y = paste0("PC2: ", round(varianza_explicada[2, 2] * 100, 2), "%"),
       title = "Componentes Principales - Estudiantes por Calificación") +
```

```
theme_minimal() +
theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14))
```

Componentes Principales – Estudiantes por Calificación

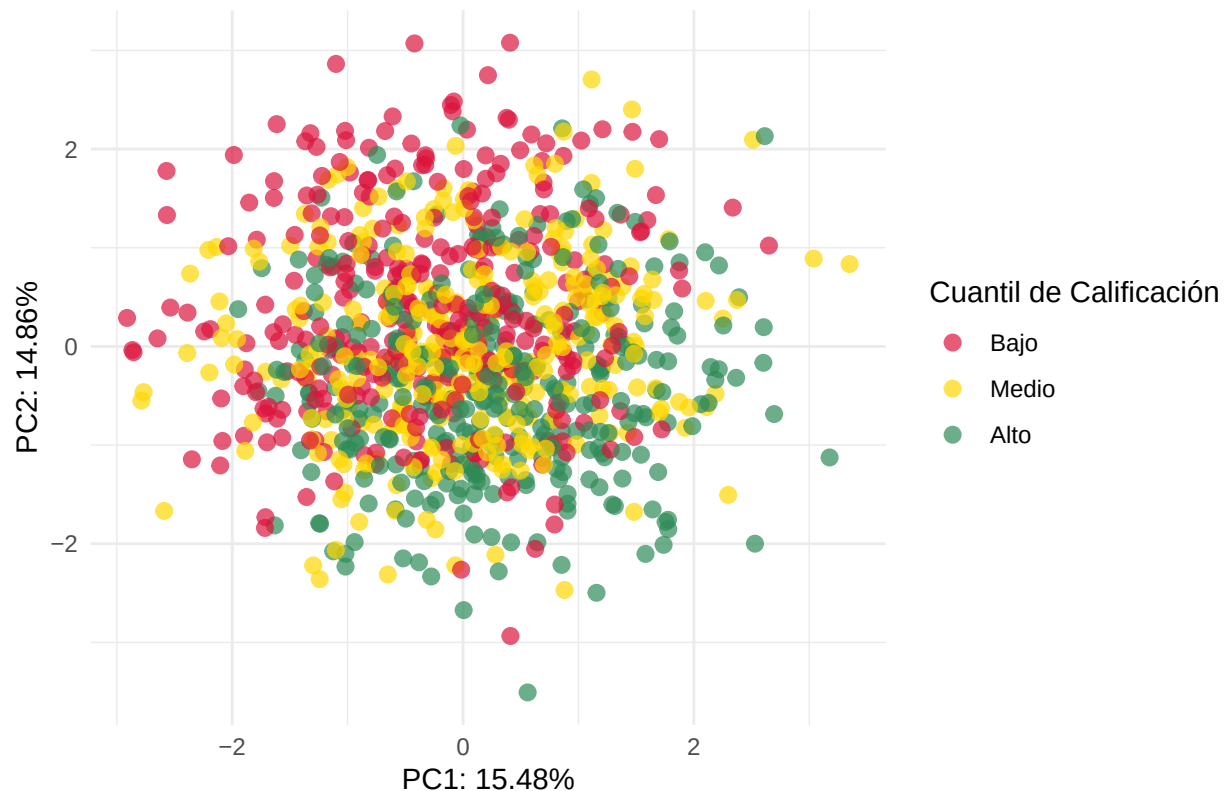


Notamos que los que se acercan al rojo están en la esquina superior izquierda, los más verdes parecen concentrarse en la esquina inferior derecha Y en medio tienes a calificaciones entre 60 y 80. Vamos a colorear de otra forma para que quede ilustrado de una mejor manera.

```
scores_pca <- as.data.frame(pca_modelo$x)
scores_pca <- scores_pca %>%
  bind_cols(exam_score = df$exam_score) %>%
  mutate(exam_score_cuantil = cut(exam_score,
                                  breaks = quantile(exam_score, probs = seq(0, 1, 1/3)),
                                  include.lowest = TRUE,
                                  labels = c("Bajo", "Medio", "Alto")))

ggplot(scores_pca, aes(x = PC1, y = PC2, color = exam_score_cuantil)) +
  geom_point(size = 2.5, alpha = 0.7) +
  scale_color_manual(
    values = c("#DC143C", "#FFD700", "#2E8B57"),
    name = "Cuantil de Calificación"
  ) +
  labs(x = paste0("PC1: ", round(varianza_explicada[2, 1] * 100, 2), "%"),
       y = paste0("PC2: ", round(varianza_explicada[2, 2] * 100, 2), "%"),
       title = "Componentes Principales – Estudiantes por Cuantil de Calificación") +
  theme_minimal() +
  theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 14))
```

Componentes Principales – Estudiantes por Cuantil de Calificación



Si bien, ayuda a visualizar que los estudiantes se agrupan por color (calificación baja arriba, alta abajo, etc) no se agrupan de una manera suficientemente tajante a comparación de los ejemplos que vimos en clase. Veamos la correlación entre las componentes y el resultado de examen.

```
correlaciones_pca <- scores_pca %>%
  summarise(
    PC1 = cor(PC1, exam_score),
    PC2 = cor(PC2, exam_score),
    PC3 = cor(PC3, exam_score),
    PC4 = cor(PC4, exam_score),
    PC5 = cor(PC5, exam_score)
  ) %>%
  pivot_longer(everything(), names_to = "Componente", values_to = "Correlación")

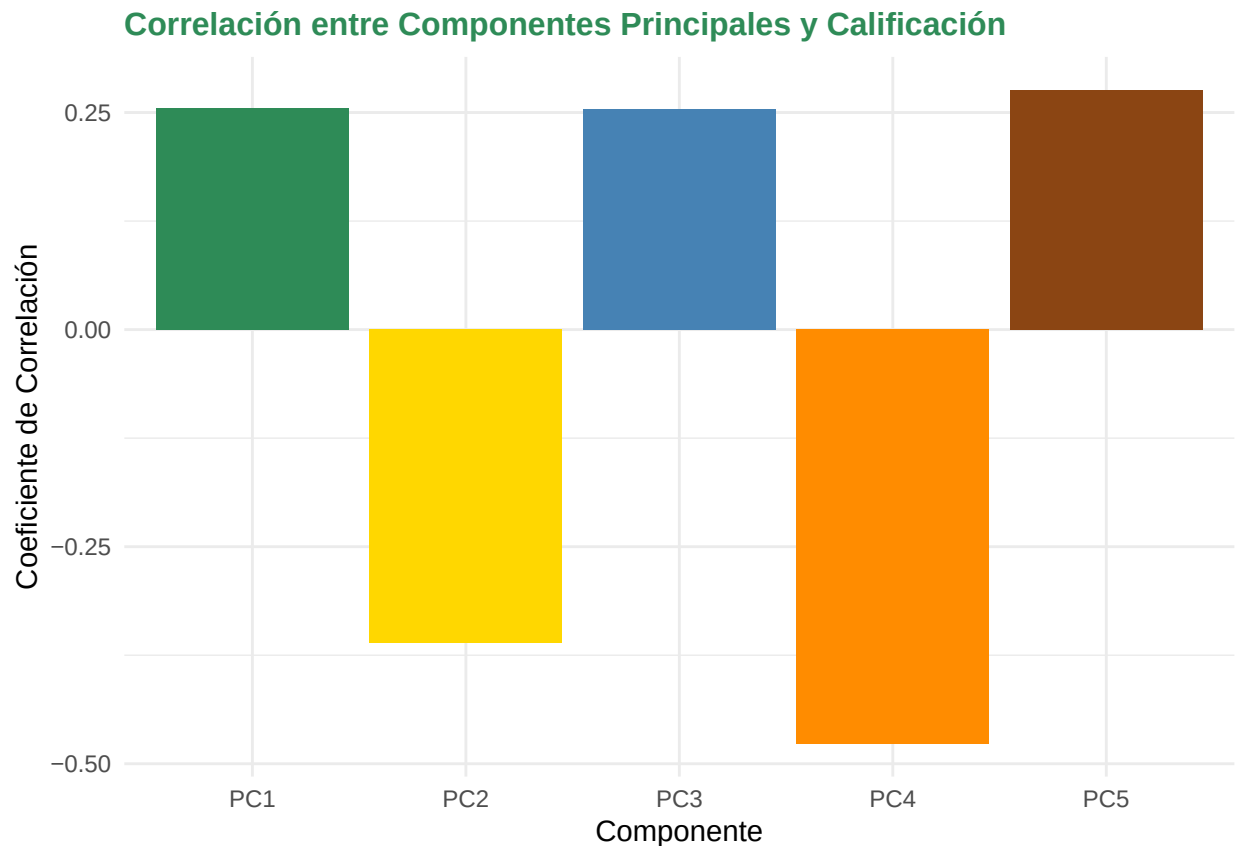
print("Correlación entre componentes y calificación:")
```

```
## [1] "Correlación entre componentes y calificación:"
```

```
print(correlaciones_pca)
```

```
## # A tibble: 5 x 2
##   Componente Correlación
##   <chr>         <dbl>
## 1 PC1           0.256
## 2 PC2          -0.361
## 3 PC3           0.254
## 4 PC4          -0.477
## 5 PC5           0.276
```

```
ggplot(correlaciones_pca, aes(x = Componente, y = Correlación, fill = Componente)) +
  geom_col() +
  scale_fill_manual(values = c("#2E8B57", "#FFD700", "#4682B4", "#FF8C00", "#8B4513")) +
  labs(title = "Correlación entre Componentes Principales y Calificación",
       y = "Coeficiente de Correlación") +
  theme_minimal() +
  theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 12),
        legend.position = "none")
```



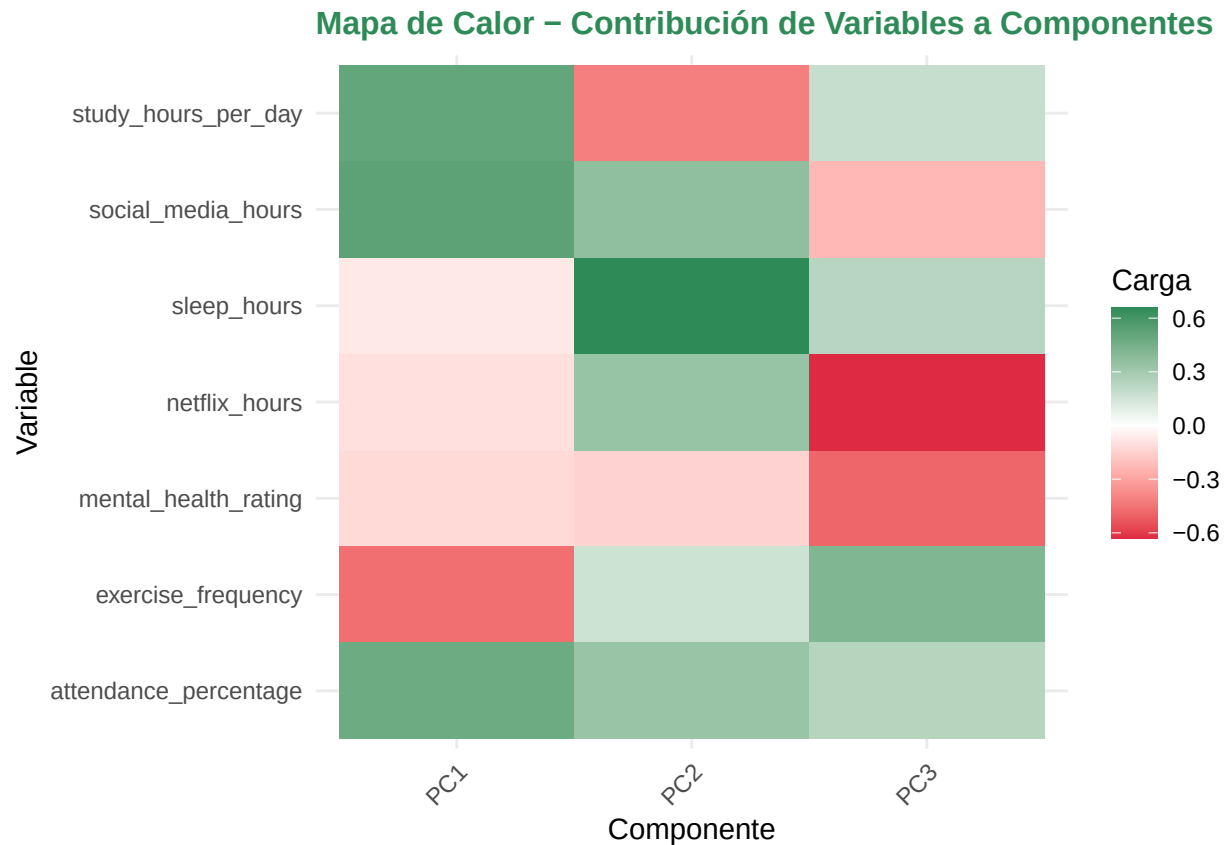
Y ahora las cargas de las componentes.

```
cargas <- as.data.frame(pca_modelo$rotation)

cargas_importantes <- cargas %>%
  select(PC1, PC2, PC3) %>%
  rownames_to_column("Variable") %>%
  pivot_longer(cols = c(PC1, PC2, PC3), names_to = "Componente", values_to = "Carga")

ggplot(cargas_importantes, aes(x = Componente, y = Variable, fill = Carga)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "#DC143C",
    mid = "#FFFFFF",
    high = "#2E8B57",
    midpoint = 0,
    name = "Carga"
  ) +
```

```
labs(title = "Mapa de Calor - Contribución de Variables a Componentes") +
theme_minimal() +
theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 12),
      axis.text.x = element_text(angle = 45, hjust = 1))
```



```
cargas_top <- cargas_importantes %>%
  group_by(Componente) %>%
  slice_max(abs(Carga), n = 5) %>%
  ungroup()

print("Variables con mayor contribución a cada componente:")

## [1] "Variables con mayor contribución a cada componente:"

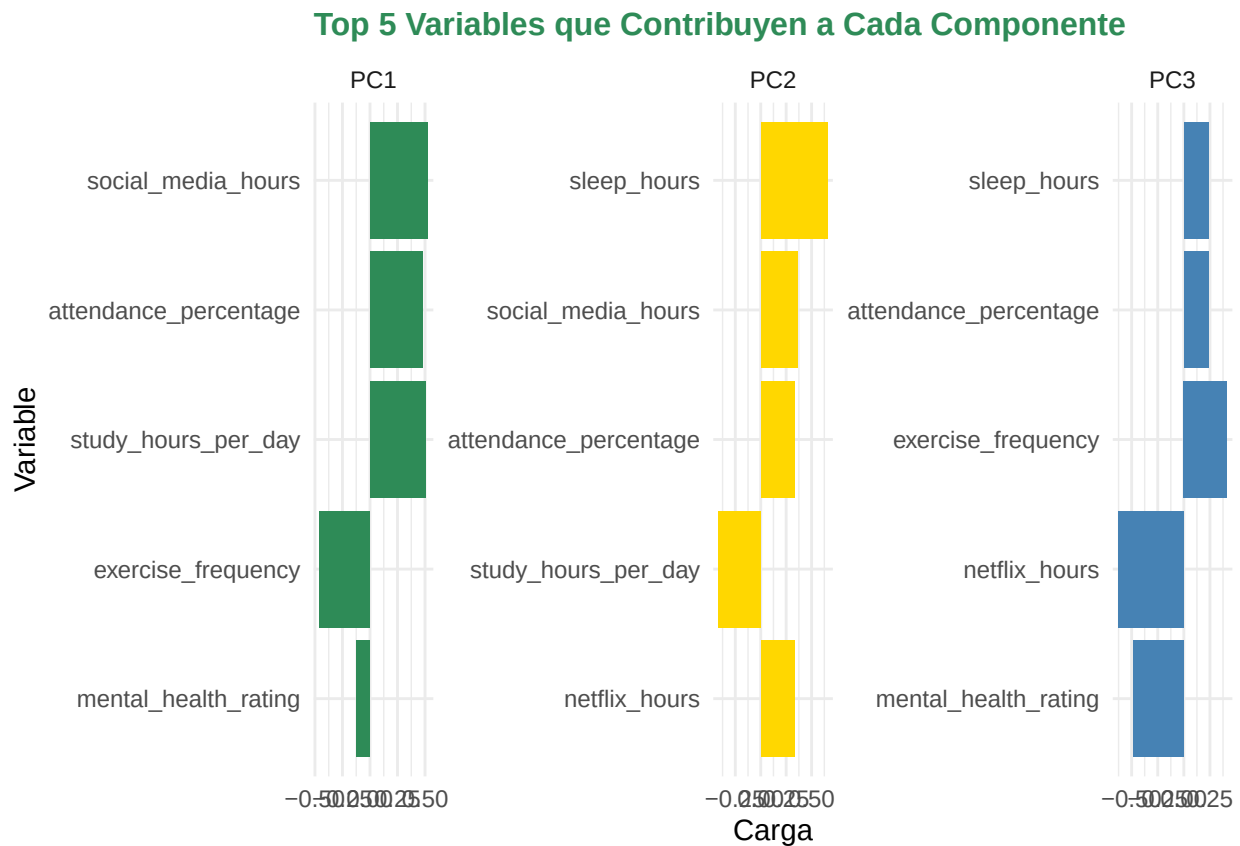
print(cargas_top)

## # A tibble: 15 x 3
##   Variable      Componente  Carga
##   <chr>         <chr>      <dbl>
## 1 social_media_hours PC1         0.526
## 2 study_hours_per_day PC1         0.504
## 3 attendance_percentage PC1         0.475
## 4 exercise_frequency PC1        -0.462
## 5 mental_health_rating PC1        -0.123
## 6 sleep_hours       PC2         0.655
## 7 study_hours_per_day PC2        -0.413
## 8 social_media_hours PC2         0.361
```



```
## 9 netflix_hours          PC2          0.334
## 10 attendance_percentage PC2          0.333
## 11 netflix_hours         PC3         -0.628
## 12 mental_health_rating  PC3         -0.487
## 13 exercise_frequency    PC3          0.413
## 14 attendance_percentage PC3          0.236
## 15 sleep_hours           PC3          0.234

ggplot(cargas_top, aes(x = reorder(Variable, Carga), y = Carga, fill = Componente)) +
  geom_col() +
  facet_wrap(~Componente, scales = "free") +
  coord_flip() +
  scale_fill_manual(values = c("#2E8B57", "#FFD700", "#4682B4")) +
  labs(x = "Variable",
       y = "Carga",
       title = "Top 5 Variables que Contribuyen a Cada Componente") +
  theme_minimal() +
  theme(plot.title = element_text(color = "#2E8B57", face = "bold", size = 12),
        legend.position = "none")
```



```
print(paste("Número total de componentes:", num_componentes))

## [1] "Número total de componentes: 7"

print(paste("Varianza total explicada por 2 componentes:",
            round(sum(varianza_explicada[2, 1:2]) * 100, 2), "%"))

## [1] "Varianza total explicada por 2 componentes: 30.34 %"
```

```
print(paste("Varianza total explicada por 5 componentes:",  
           round(sum(varianza_explicada[2, 1:5]) * 100, 2), "%"))
```

```
## [1] "Varianza total explicada por 5 componentes: 73.08 %"
```

```
print(paste("Varianza total explicada por 5 componentes:",  
           round(sum(varianza_explicada[2, 1:5]) * 100, 2), "%"))
```

```
## [1] "Varianza total explicada por 5 componentes: 73.08 %"
```

```
print(paste("Mejor correlación con calificación:",  
           round(max(abs(correlaciones_pca$Correlación)), 3)))
```

```
## [1] "Mejor correlación con calificación: 0.477"
```

En conclusión, fue una buena idea eliminar las categóricas pues ayudó a explicar mejor la varianza. No obstante, no fue suficiente para agrupar de manera satisfactoria.